2.3. Capacitance, resistance & current

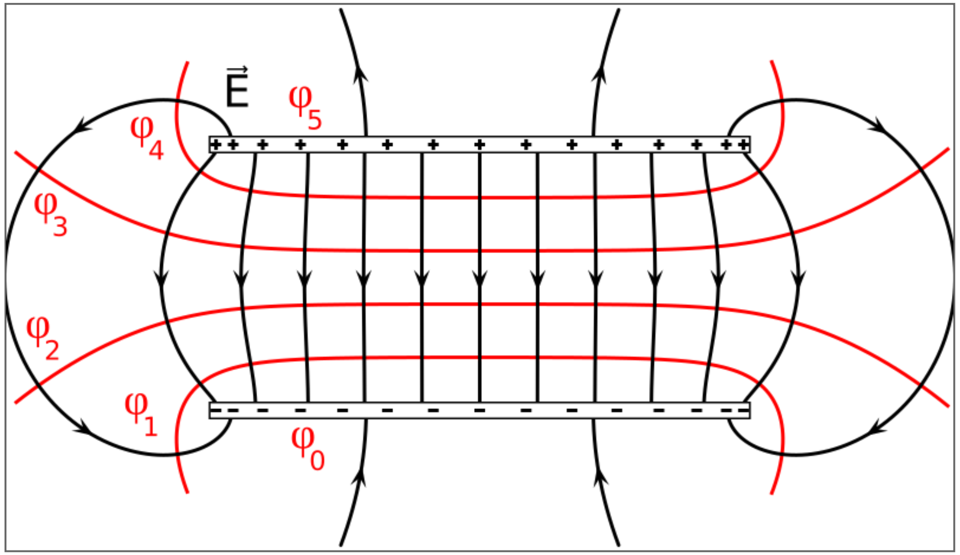
`el25 - apple power`

How can we store electric energy?

- using circuit elements such as:
  - capacitors (separated charges)
  - electric batteries (electrochemical voltage source)
- transition from **static electricity to flow of charges**
- associated physical concepts: dielectrics, electric power, current, resistance, & Ohm's law
- **disclaimer**: simplify notation for voltage to $V = V_{BA} = V_B - V_A$

Capacitors - Basic concept

- two plates with opposite charge produce an electric field
- example for parallel plates:



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Capacitors - Relation of V and d at parallel plates

`es27`

- two parallel plates of area A separated by distance d
- voltage source connected
- $\rightarrow$ plates accumulate charge Q with equal magnitude but opposite sign

What happens if we change the distance? (if $Q = \mathrm{const}$)

- **voltage increases with distance** (until breakdown voltage reached)

$$V = V_B - V_A = -\int_A^B \vec{\mathbf{E}} d\vec{\mathbf{l}} = +E \int_A^B dl = Ed = \frac{Q}{\epsilon_0 A} d$$

Notes:

- electric field for two parallel planes: $E = \frac{Q}{\epsilon_0 A}$

- choose path from "-" to "+": $d\vec{\mathbf{l}}$ antiparallel to $\vec{\mathbf{E}}$ (180°)

Capacitance C

- **fundamental relation**: $Q = CV$
- **capacitance**:
  - proportionality constant $C = \frac{Q}{V}$
  - unit: **farad** [F] = [C/V], typically capacitors in picofarad to microfarad range
  - capacitance determined by **geometry**: size, shape, relative position of plates
- determining capacitance **analytically** for an uniform electric field $E = \frac{Q}{\epsilon_0 A}$:

$$V = Ed = \frac{Qd}{\epsilon_0 A}$$

$$\frac{V}{Q} = \frac{d}{\epsilon_0 A}$$

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d}$$

Storing electric energy

`sim: C, Q, E, U`

- **conservation of energy**: the work W required to charge the capacitor is equal to the electric energy stored in the capacitor U
- work required to move small amount of charge in presence of potential difference: $dW = -V\,dq$
- integrating over the entire charge Q and with $V = \frac{q}{C}$, we get:

$$W = -\int_0^Q V\,dq = -\frac{1}{C}\int_0^Q q\,dq = -\frac{1}{2}\frac{Q^2}{C}$$

- $\rightarrow$ energy U "stored" is:

$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$$

Dielectrics

`es21`

- so far: **vacuum between plates** (ϵ_0)
- goal: **store more energy at higher voltage safely**
- **dielectrics are insulating materials**
- inserting a dielectric allows higher breakdown voltage → smaller gaps & more compact capacitors
- **capacitance increases by factor K:** $C = K\, C_0$
- material permittivity: $\epsilon = K\, \epsilon_0$
- parallel-plate capacitor: $C = \frac{\epsilon A}{d}$

Dielectrics: fixed Q vs. fixed V

Case 1: isolated capacitor (fixed Q)

- no battery → charge cannot change
- inserting dielectric:
  - electric field **decreases**
  - voltage **decreases** ($V = \frac{Q}{C}$, $C \uparrow$)
  - stored energy **decreases**

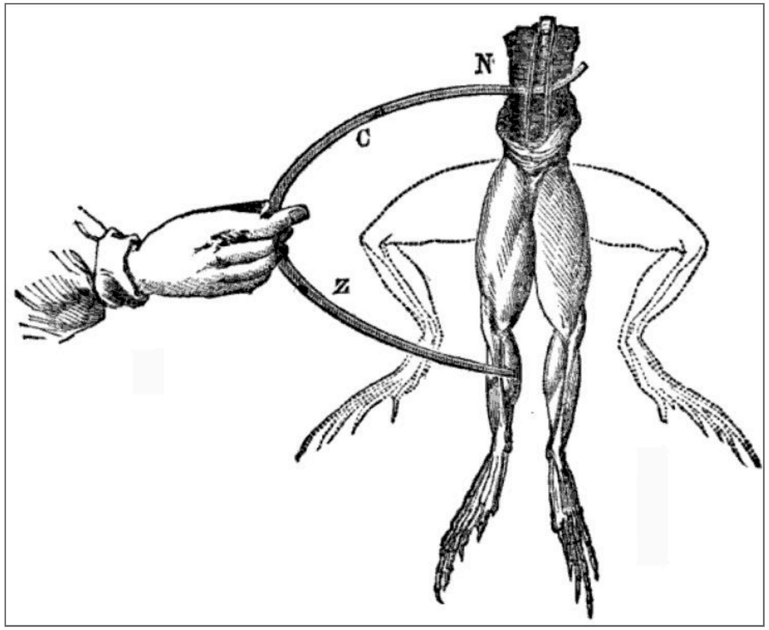
Case 2: connected to battery (fixed V)

- battery keeps voltage constant
- inserting dielectric:
  - electric field **unchanged** ($E = V/d$)
  - additional charge flows → $Q \uparrow$
  - stored energy **increases**

Summary: dielectric always increases C, but what changes depends on constraints

History of electric battery: Galvani vs. Volta

- Luigi Galvani (1737-1798) connected a cupper and iron wire to a frog leg and saw muscle contraction
- $\rightarrow$ interpreted as *life-force*
- Alessandro Volta (1745-1827) disagreed and realized the potential (pun intended) of dissimilar metal
- $\rightarrow$ combined cells of zinc & silver soaked in salt solution to form a *battery* of cells



*from **wikipedia**, public domain*

Concept of electric battery

`el23 & revisist el25`

- battery consists of:
    - **electrodes** - metal rods connected to terminals:
        - **cathode** - negative electrode e.g. Pb (lead)
        - **anode** - positive electrode e.g. Pb02 (lead dioxide)
    - **electrolyte** e.g. sulfuric acid, apple, or frog leg
- oversimplified reaction: Pb + acid $\rightarrow$ net effect of accumulation of electrons in cathode
- if connected, battery provides a voltage that drives current in a circuit
- **no charges are generated, merely separated, obeying the laws of conservation**

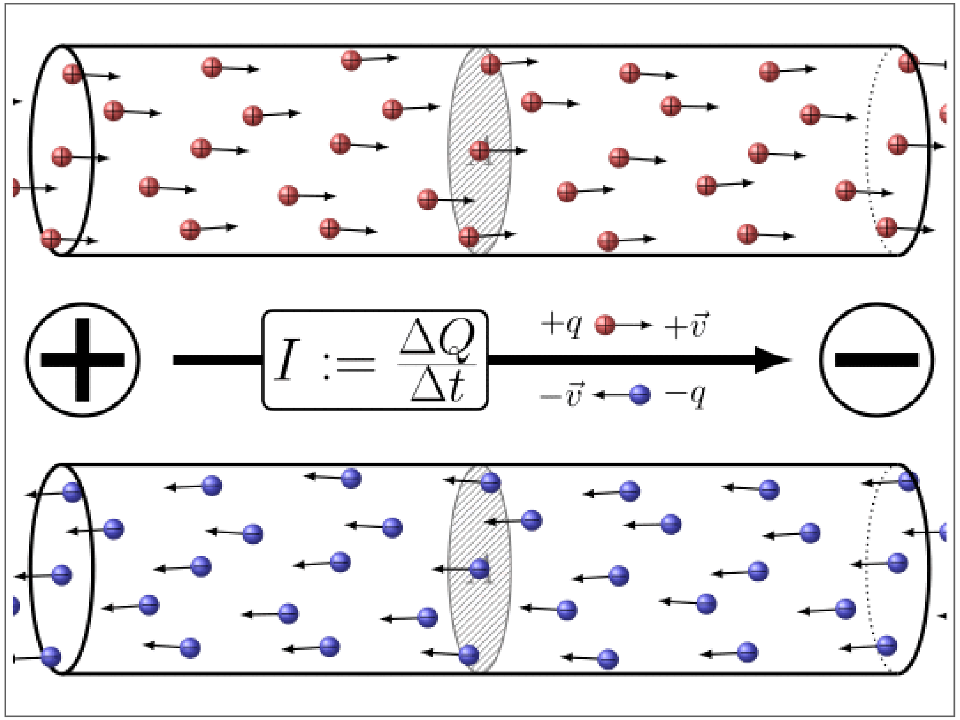
Start moving: electric current, resistance & ohm's law

`es15`

Time to leave electrostatics, i.e. resting charges, and consider moving charges

Electric current

- average current defined as $\bar{I} = \frac{\Delta Q}{\Delta t}$
- instantaneous current defined as $I = \frac{dQ}{dt}$
- **unit** *ampere* [A]=[C/s] in recognition of André Ampère (1775-1836)
- **conservation of charge**: current is constant throughout a continuous conductor
- **flow direction**:
  - conventional current flows from positive to negative (Franklin), while electrons move from negative to positive (physics)
  - directionality (usually) non-critical/yield equal results (exception e.g. Hall effect)


$+q$
$+\vec{v}$
$I := \frac{\Delta Q}{\Delta t}$
$-\vec{v}$
$-q$

Ohm’s law & resistors

- **Ohm's law** relates voltage, current, and resistance: $V = I\,R \leftrightarrow I = \frac{V}{R}$
- **resistance:**
    - proportionality constant which quantifies the **hindrance of electron flow**
    - unit of resistance: **Ohm** ($\Omega = \frac{V}{A}$) in recognition of Georg Simon Ohm (1787-1854)
- Ohmic resistors have a constant R, while non-ohmic resistors change with conditions such as temperature

Resistivity & conductivity

`el32 + el04`

- **resistance** in a uniform wire **depends on**:
    - cross-sectional area A
    - wire length l,
    - material used, i.e. resistivity ρ in [Ω m]:

$$R = \rho \frac{l}{A}$$

- alternative material property: conductivity: $\sigma = \frac{1}{\rho}$

Temperature dependency of resistivity

`el03`

- **temperature dependence of resistivity** can be approximated as

$$\rho(T) = \rho_0 \left(1 + \alpha \left(T - T_0\right)\right)$$

    - α being the material-specific **temperature coefficient** of resistivity
    - **negative temperature coefficients (NTC)**, i.e. lower resistance when heated, such as semiconductor which have more free electrons available at higher temperatures
    - **positive temperature coefficients (PTC)**, i.e. higher resistance when heated, such as many metals as the higher temperature increases the likelihood of atom-electron collision

Electric power

- **electric circuits transmit electric energy** $\rightarrow$ amount of electric power P delivered is therefore of interest (at the very least to you energy provider)
- **electric power is the energy per unit time**: $P = \frac{dU}{dt}$
- unit: **Watt** [W]=[J/s]
- using $V = \frac{U}{q} \rightarrow dU = V dq$ as well as $I = \frac{dq}{dt}$

$$P = \frac{dU}{dt} = \frac{dq}{dt} V$$

$$P = VI$$

- applying Ohm's law $V = RI$, we can extend this to:

$$P = VI = I^2 R = \frac{V^2}{R}$$

Why current density?

`el07`

- current I tells us **how much charge per second** passes through a surface
- but I alone does **not** tell us how this flow is distributed
- the same current can flow through:
    - a **thick** wire
    - a **thin** wire
    - a conductor with **non-uniform flow**
- therefore we need a **local quantity**
- **current density** describes charge flow **per unit area** and **at each point**
- it tells us where charge flow is strong, weak, or non-uniform

Current density

- current density $\vec{\mathbf{j}}$ describes the **local flow of charge**
- its direction is the direction of **positive charge flow**
- for uniform flow through a flat cross-section:

$$j = \frac{I}{A} \qquad \leftrightarrow \qquad I = jA$$

- for non-uniform current flow:

$$I = \int \vec{\mathbf{j}} \cdot d\vec{\mathbf{A}}$$

- thus:
    - I is the **total current** through a surface

- $\vec{\mathbf{j}}$ tells us **locally** how much current passes through unit area

Drift speed

- macroscopic perspective: electricity moves with the speed of light
- microscopic perspective:
    - electrons collide with lattice
    - electrons move with an **average drift speed** $v_d \approx 0.05$ mm/s
- relate drift speed to (macroscopic) current via number of free electron per unit volume $n = \frac{N}{Al}$:

$$I = \frac{\Delta Q}{\Delta t} = \frac{-eN}{\Delta t} = \frac{-enAl}{\Delta t} = \frac{-enAv_d\Delta t}{\Delta t} = -enAv_d$$

$$j = \frac{I}{A} = -env_d$$

- minus sign indicates that electrons drift in opposite direction to macroscopic current

Microscopic view on Ohm's law

- macroscopic Ohm's law:

$$V = RI$$

- for a homogeneous conductor: $R = \rho \frac{l}{A}, \quad V = El, \quad I = jA$
- insert into Ohm's law:

$$El = \rho \frac{l}{A} jA$$

$$E = \rho j$$

- microscopic form of Ohm's law:

$$\vec{\mathbf{E}} = \rho \vec{\mathbf{j}} \qquad \leftrightarrow \qquad \vec{\mathbf{j}} = \frac{1}{\rho} \vec{\mathbf{E}}$$

- interpretation:
    - $\vec{\mathbf{E}}$ drives the charges $\rightarrow$ proxy for V

- $\vec{\mathbf{j}}$ describes the resulting local current flow $\rightarrow$ proxy for I
- ρ is a material property $\rightarrow$ proxy for R

2.3. Capacitance, resistance & current

`el33 - TOLMAN-Experiment`

Experiment shows:

- electrons have mass & interia (start / stop)
- sign of current $\rightarrow$ carriers are negative $\rightarrow$ electrons

Calculating the voltage:

$$F_C = F_Z$$

$$eE = m_e\,\omega^2 r \quad \Rightarrow \quad E = \frac{m_e}{e}\,\omega^2 r$$

$$V = \int_0^R E(r)\,dr = \int_0^R \frac{m_e}{e}\,\omega^2 r\,dr = \frac{m_e}{2e}\,\omega^2 R^2$$